

PCAP-Lab1- Submission

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Q1)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q1.c -o q1
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q1
Thread - 3
Thread - 14
Thread - 4
Thread - 12
Thread - 8
Thread - 9
Thread - 10
Thread - 13
Thread - 2
Thread - 11
Thread - 1
Thread - 5
Thread - 6
Thread - 15
Thread - 7
Thread - 0
Total Number of threads: 16
```

Q2)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q2.c -o q2
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q2
Enter number of threads:
3
Thread ID - 0
Thread ID - 1
Thread ID - 2
```

Q3)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q3.c -o q3
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q3
Row - 1 => Thread - 1
Row - 0 => Thread - 0
Row - 3 => Thread - 3
Row - 2 => Thread - 2
Row - 4 => Thread - 4
MATRIX:
1      2      3      4      5
6      7      8      9      10
11     12     13     14     15
16     17     18     19     20
21     22     23     24     25
```

Q4)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q4.c -o q4
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q4
Enter size of the arrays:
4
Enter array1:
1 43 897 23
Enter array2:
43 901 0 1
arr[3]=24 => Thread - 3
arr[2]=897 => Thread - 2
arr[1]=944 => Thread - 1
arr[0]=44 => Thread - 0
```

Q5)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q5.c -o q5
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q5
Enter matrix dimensions (Rows Columns): 3
3

Enter elements for Matrix A (3x3):
A[0][0]: 1 2 3 4 5
A[0][1]: A[0][2]: A[1][0]: A[1][1]: A[1][2]: 6 7 8 9
A[2][0]: A[2][1]: A[2][2]:
Enter elements for Matrix B (3x3):
B[0][0]: 67 92 2 9 0
B[0][1]: B[0][2]: B[1][0]: B[1][1]: B[1][2]: 2 1 1 1 1
B[2][0]: B[2][1]: B[2][2]:
--- Parallel Matrix Addition ---
Row 2 computed by Thread 2
Row 1 computed by Thread 1
Row 0 computed by Thread 0

--- Resultant Matrix C (3x3) ---
68      94      5
13      5      8
8       9      10
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ |
```

Q6)

```
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ gcc -fopenmp q6.c -o q6
dhanunjay@Dhanunjay:~/Dhanunjay_240962018/Lab1$ ./q6
--- Array Sum Performance Comparison ---
Array Size      : 50000000 elements
Threads Used    : 16

Serial Sum      : 50000000
Serial Time     : 0.073894 seconds

Parallel Sum    : 50000000
Parallel Time   : 0.017825 seconds

Speedup : 4.15x faster
```